Data Science in Cyber — Final Project Report

Critical Reproduction & Evaluation: Arcos-Argudo et al. (2025) on NSL-KDD

This report presents an independent reproduction and critical evaluation of Arcos-Argudo et al. (2025), who compare classical machine learning classifiers and an Autoencoder + Logistic Regression (AE+LR) hybrid for binary intrusion detection on NSL-KDD.
We reproduced all author NSL-KDD metrics within ±0.02 using their replication package, then rebuilt the pipeline from raw KDDTrain+.txt / KDDTest+.txt with leakage-aware preprocessing.
Key findings: LDA matches AE+LR on F1 at far lower training cost; Random Forest (our extension) achieves the highest AUC (0.960); SMOTE yields modest aggregate gains but does not fix rare-attack detection (R2L recall ~8%); Naïve Bayes confirms extreme precision/recall asymmetry; and the author notebook's train+test concat encoding contradicts a strict leakage-free claim.
Recommendation: adopt the paper's metric framework (especially FAR) for tabular IDS benchmarks, but do not treat AE+LR superiority or high AUC as evidence of deployable detection without per-family evaluation.

1. Summary of the Source

Problem. Intrusion detection systems (IDS) must distinguish malicious from benign network traffic under class imbalance and asymmetric error costs — false negatives miss breaches; false positives burden analysts.
Source. Arcos-Argudo et al. (2025), Algorithms 18(12), 749, compare Naïve Bayes, Logistic Regression, Linear Discriminant Analysis, and AE+LR on NSL-KDD (binary Normal vs Attack) and CICIDS2017. We focus on NSL-KDD per assignment scope.
Dataset. Official split: KDDTrain+.txt (125,973 records) and KDDTest+.txt (22,544 records); 41 features plus label and difficulty (dropped). No timestamps — connection-level features only.
Methodology. One-hot encoding of protocol_type, service, flag; min-max scaling fit on train; optional SMOTE on train only; fixed random_state=42. AE: Dense(64)→32→64, 20 epochs; LR on 32-dim embeddings. Metrics: Accuracy, Precision, Recall, F1, FAR, ROC-AUC.
Importance. The paper claims AE+LR achieves the highest AUC (~0.904), SMOTE helps modestly, preprocessing is leakage-free, and the replication package enables byte-level reproducibility — claims we test with independent evidence.

2. Critical Evaluation

We evaluate seven author claims (C1–C7) using results/baseline_reproduction.csv (author pipeline) and results/experiment_metrics.csv (our train-only pipeline from raw data).
C1 (AE+LR best AUC/F1): PARTIALLY SUPPORTED. Author baseline: AE+LR AUC 0.904, F1 0.753 (top among four). Our pipeline: RF AUC 0.960, LDA F1 0.752; AE+LR below both. Hybrid advantage is pipeline-dependent.
C2 (SMOTE modest F1 gains): SUPPORTED. Author ΔF1 +0.002–0.005; ours up to +0.031 (AE+LR). Always train-only.
C3 (Leakage-free preprocessing): REJECTED for author code / SUPPORTED for ours. Author concat train+test before get_dummies introduces schema leakage. Our src/preprocessing.py fits encoders on train only.
C4 (LR/LDA strong baselines): SUPPORTED. LDA F1 within 0.002 of AE+LR in both pipelines.
C5 (NB high Prec, low Rec): SUPPORTED. Prec ~0.97–0.98, Rec ~0.15–0.24, FAR 0.65%.
C6 (FAR essential): SUPPORTED. NB accuracy 0.51 with minimal FAR but ~85% missed attacks; accuracy alone misleads.
C7 (Byte-level reproducibility): PARTIALLY SUPPORTED. All Table 4/5 metrics matched with bundled CSVs; hidden CSV ETL and library drift prevent reproduction from raw .txt alone.
Limitations. Binary collapse hides R2L/U2R failure; NSL-KDD is legacy; near-ceiling DoS/Probe metrics lack modern external validity; fixed 0.5 threshold ignores operational calibration.

3. Feature Engineering Analysis

Transformations applied (our pipeline, src/preprocessing.py):
1. Binary label: normal→0, all attacks→1 (matches paper task definition). 2. Drop constant feature num_outbound_cmds (zero variance in train). 3. Drop highly correlated numerics (|Pearson r| ≥ 0.95 on train): seven error-rate/host statistics removed. 4. One-hot encode categoricals — train-only schema (106 features) vs author concat style (119 features). 5. MinMaxScaler fit on train, applied to test. 6. SMOTE optional on train only (random_state=42, k_neighbors=5).
Redundancy. EDA and correlation analysis show clusters among dst_host_* error rates and serror/rerror features. Removing them reduces multicollinearity without large metric loss for LR/LDA.
Cybersecurity rationale. Protocol/service/flag encode attack vectors; byte and connection counts capture volume anomalies; scaling puts heterogeneous counts on comparable ranges for linear models.
Feature creation (exploratory). We evaluated derived candidates such as bytes_ratio = src_bytes / (dst_bytes + 1) for asymmetric traffic patterns; they show class separation in EDA but were not added to final models to preserve comparability with the authors' feature space.
Suggested improvements. Per-family or hierarchical labels for R2L/U2R; log-scaled byte counts; threshold tuning on validation data; audit encoding for leakage; report MCC and per-family recall alongside binary F1.

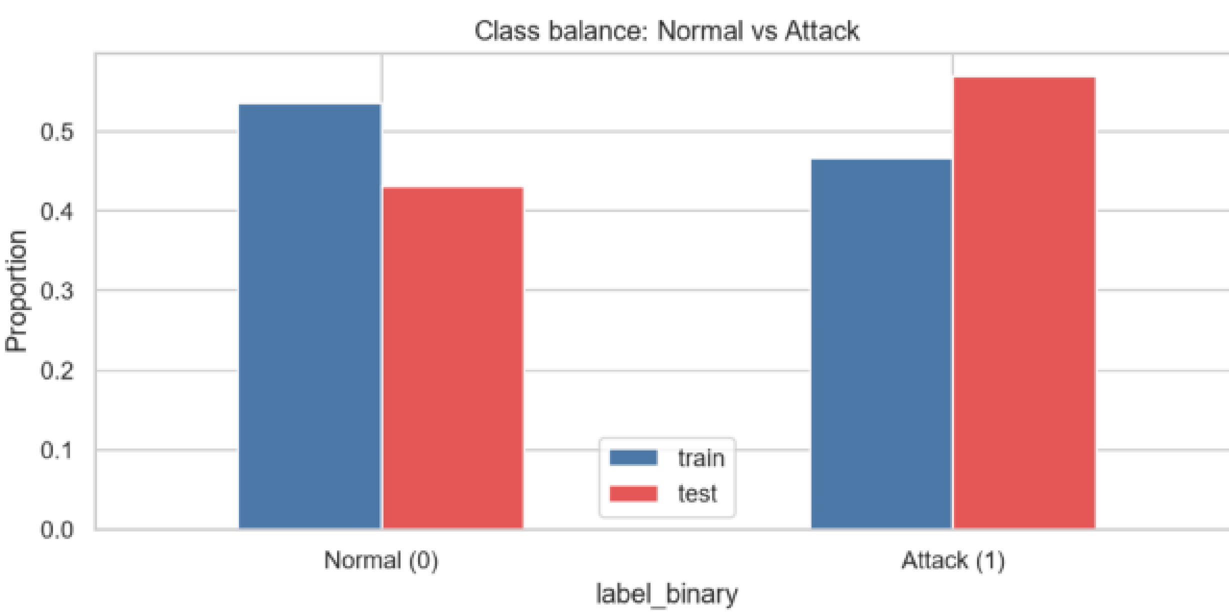
4. Reproducibility Analysis

Execution. Author baseline via scripts/run_phase1_baseline.py mirroring vendor/.../BACK-nsl-kdd-github.ipynb. Our pipeline via src/ modules and notebooks/ids_nsl_kdd_analysis.ipynb from raw NSL-KDD files.
Environment. Python 3.12.10, scikit-learn 1.9.0, TensorFlow 2.21.0, imbalanced-learn 0.14.2 (Windows 10 CPU). Paper reports TF 2.20 / sklearn 1.7.2 — minor AE+LR floating-point drift observed (F1 $\Delta \leq 0.0004$).
Reproduction outcome. All eight author model/regime combinations MATCH paper Tables 4–5 (max $\Delta \leq 0.02$).
Hidden steps. Author notebook uses pre-cleaned binary CSVs; raw KDDTrain+.txt never loaded; cleaning script absent. One row dropped per split (125,972 / 22,542 vs 125,973 / 22,544). README lists wrong notebook path.
Overall reproducibility. Functional for metric verification with bundled artifacts; not end-to-end from primary data files without reverse-engineering. Our reimplementation documents the gap and applies strict train-only encoding.

5. Experimental Results

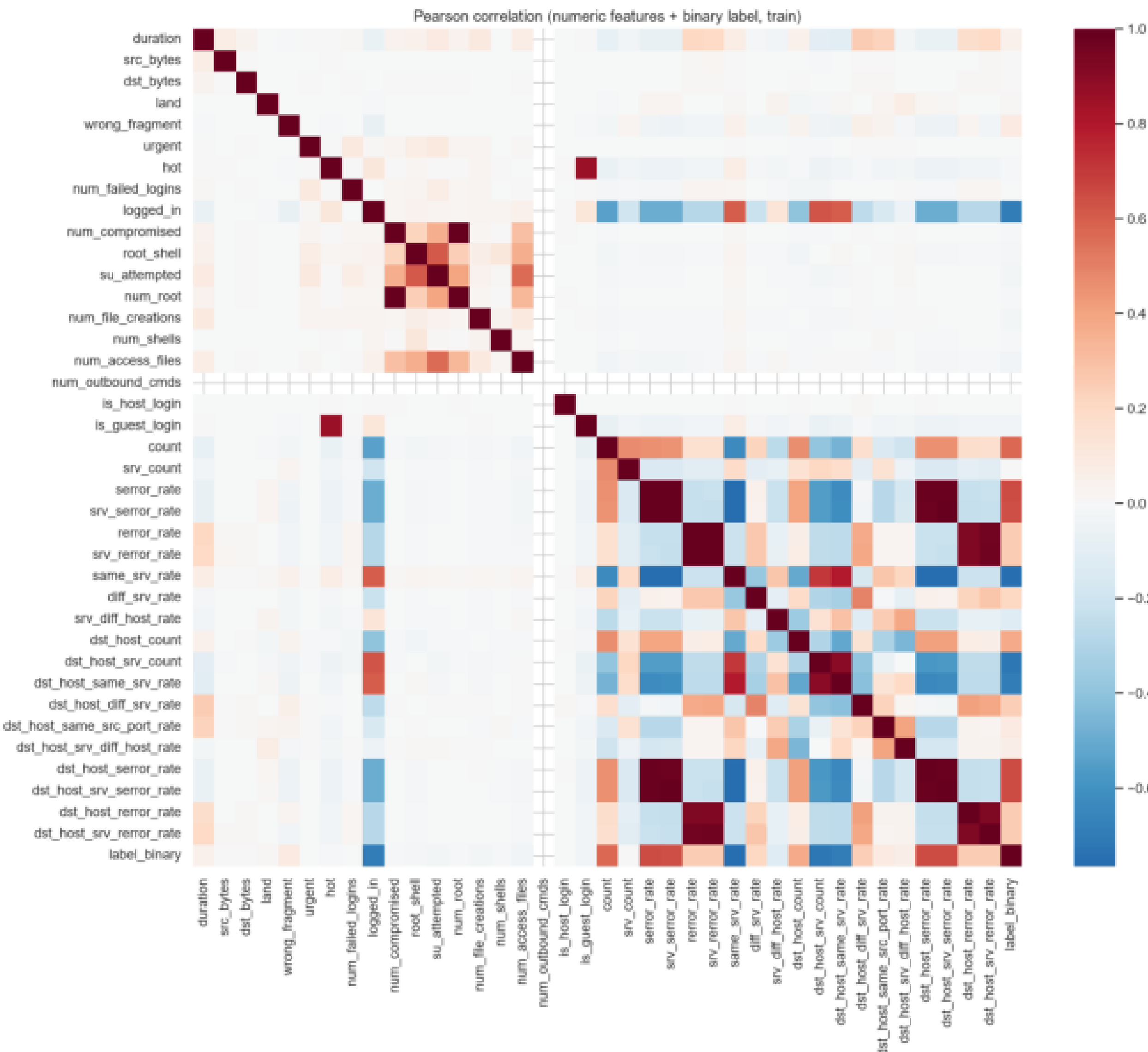
Our experiments extend the paper with Random Forest (n_estimators=100) and train-only preprocessing from raw NSL-KDD. Below: test-set metrics without SMOTE (our pipeline).
model smote accuracy precision recall f1 far auc AE+LR no 0.7439 0.9114 0.6094 0.7304 0.0783 0.8959 LDA no 0.7618 0.9242 0.6334 0.7517 0.0687 0.8601 LR no 0.7340 0.9120 0.5897 0.7162 0.0752 0.8431 NB no 0.5121 0.9679 0.1478 0.2565 0.0065 0.7944 RF no 0.7669 0.9666 0.6115 0.7491 0.0279 0.9597
SMOTE impact (ΔF1): AE+LR +0.0015, LDA +0.0030, LR +0.0056, NB +0.0000, RF +0.0129.
Error analysis (RF, no SMOTE): DoS recall 0.80, Probe 0.69, R2L 0.08, U2R 0.10. Rare subclasses such as guess_passwd and snmpguess show 0% recall under LR.
Author baseline comparison: LR and LDA within ±0.02 of author metrics; NB and AE+LR diverge due to encoding and feature-set differences (documented in results/baseline_comparison.csv).
Metric selection. We report Accuracy, Precision, Recall, F1, MCC, FAR, and AUC. Fβ is omitted by design: FAR quantifies false alarms on normal traffic, recall covers missed attacks, and LR threshold sweeps expose the operational FP/FN trade-off without fixing β. MCC summarizes performance under class skew.

Figure 1 — Class balance



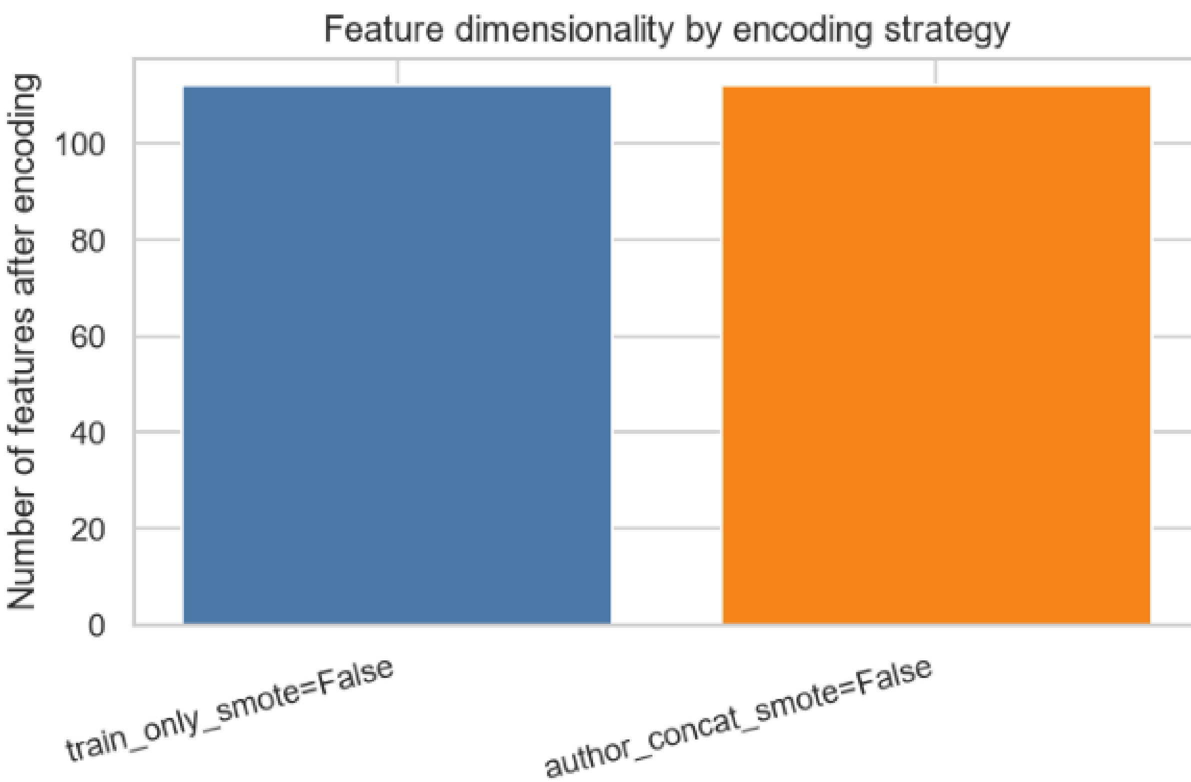
Train vs test label distribution; imbalance motivates train-only SMOTE.

Figure 2 — Feature correlation (Pearson)



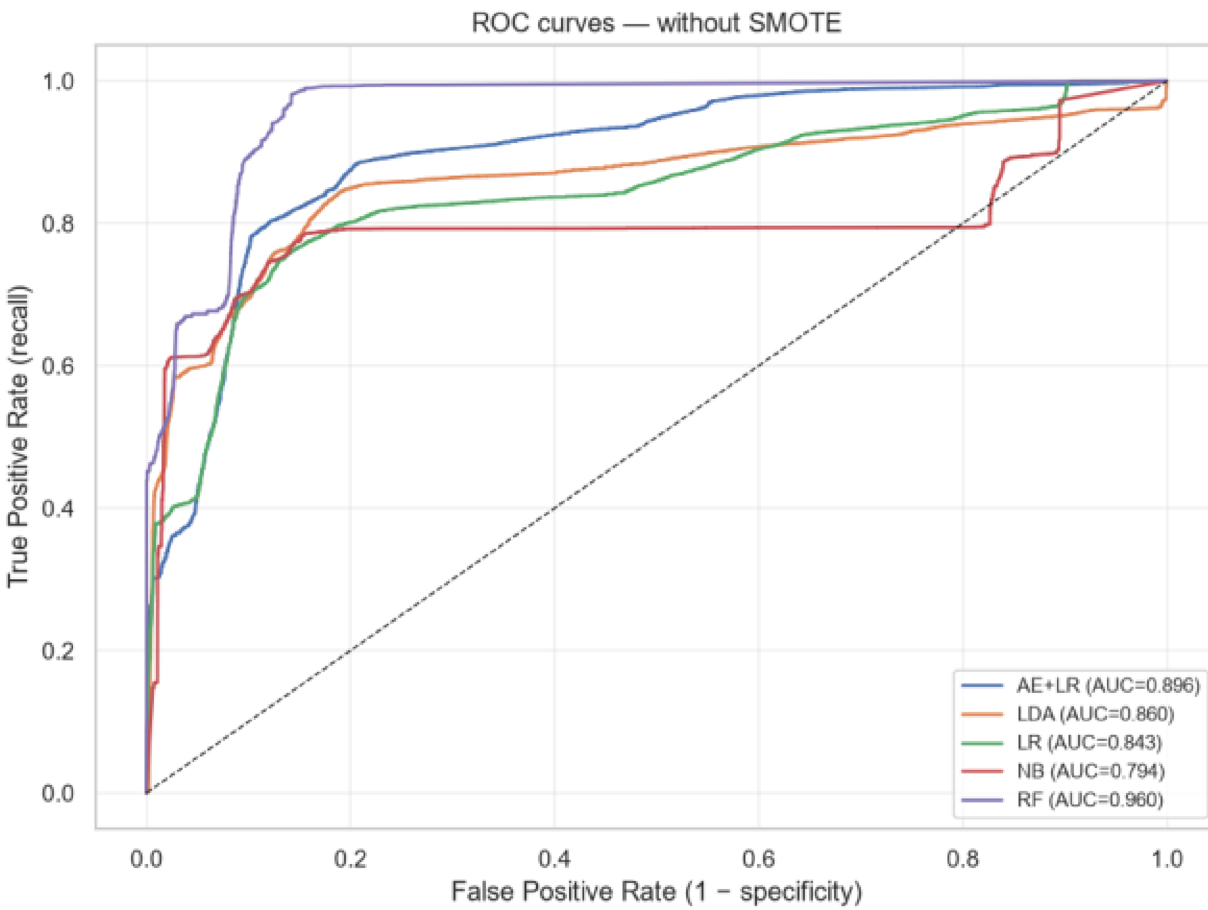
Redundant error-rate clusters motivate correlation-based feature drops.

Figure 3 — Encoding feature counts



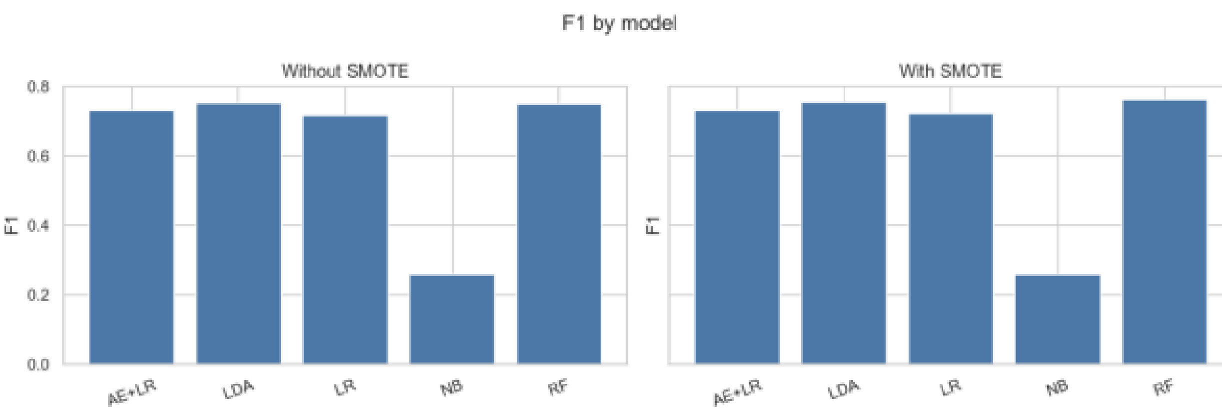
Train-only (106) vs author concat encoding (119 features).

Figure 4 — ROC curves (no SMOTE)



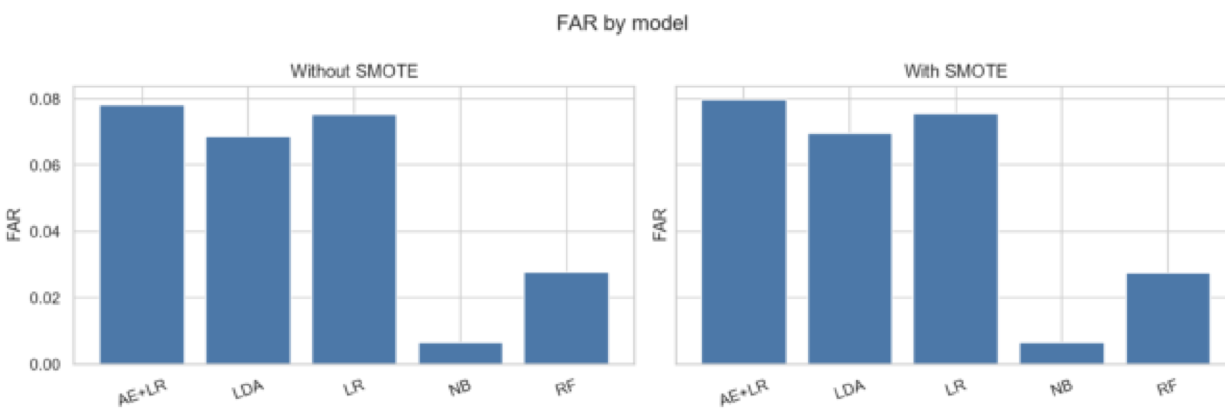
RF dominates AUC; AE+LR below RF and comparable to LDA.

Figure 5 — F1 by model



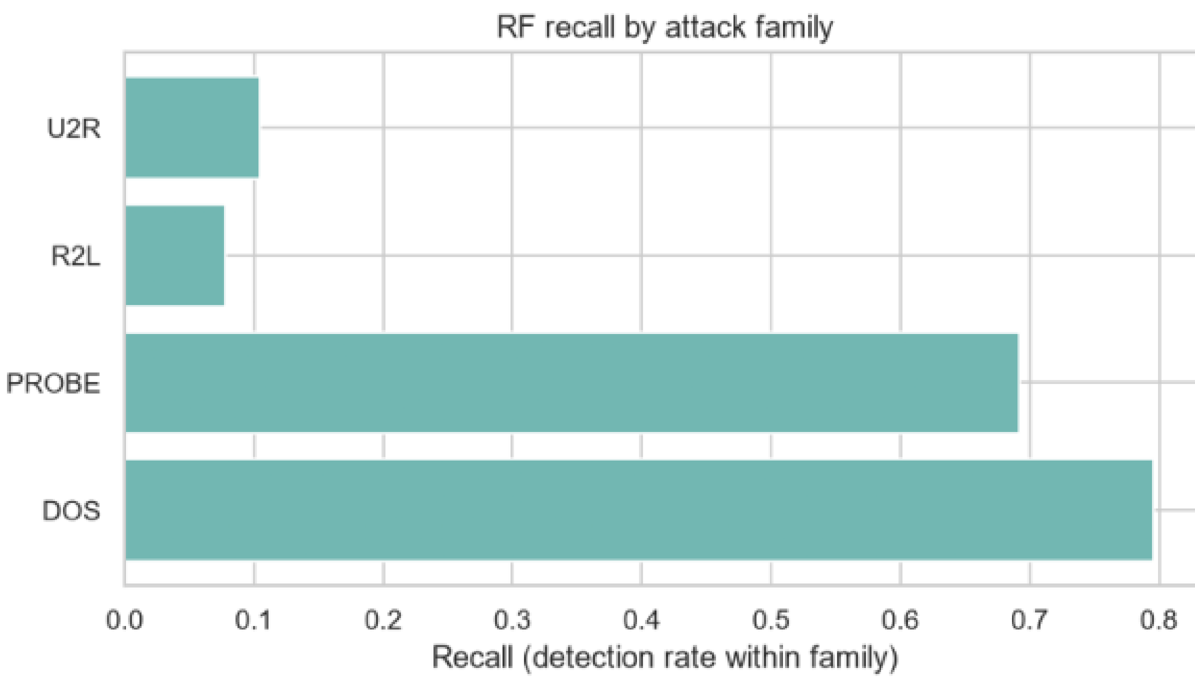
LDA best F1 among paper models; NB lowest.

Figure 6 — FAR by model



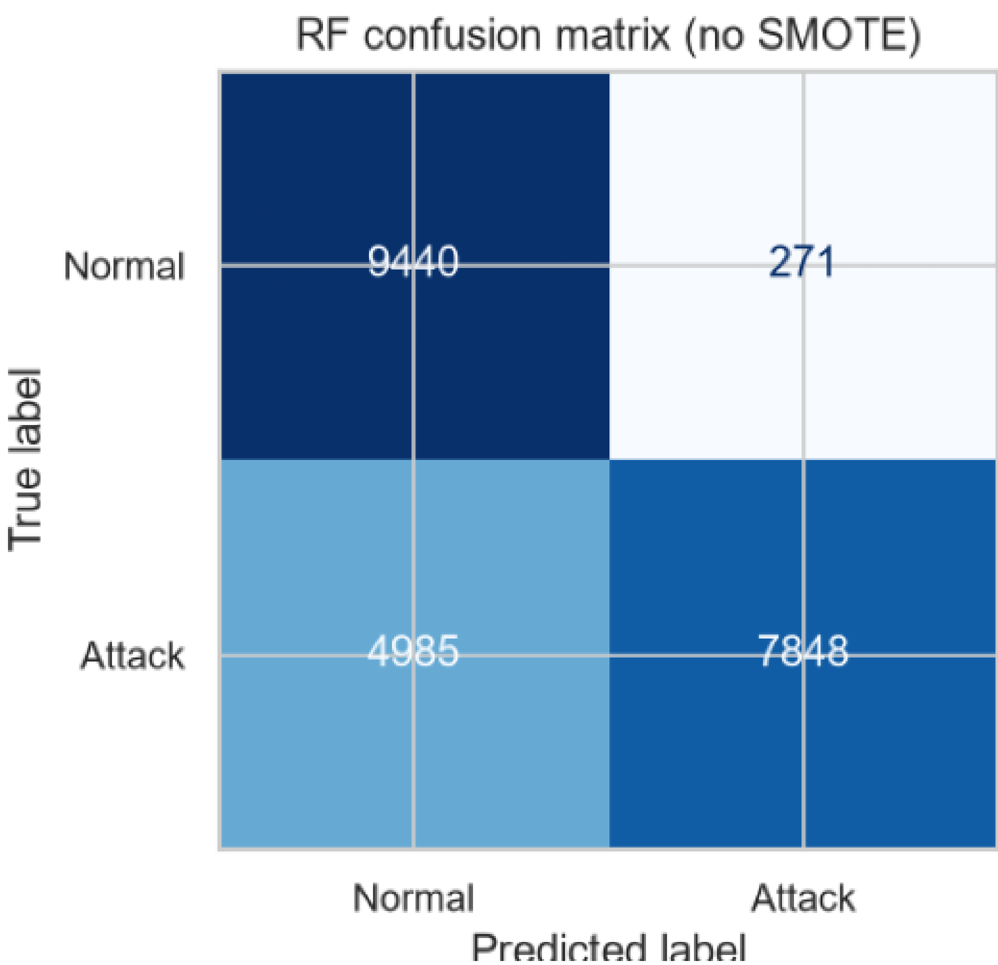
NB minimal FAR vs higher FAR for LR/AE+LR.

Figure 7 — RF recall by attack family



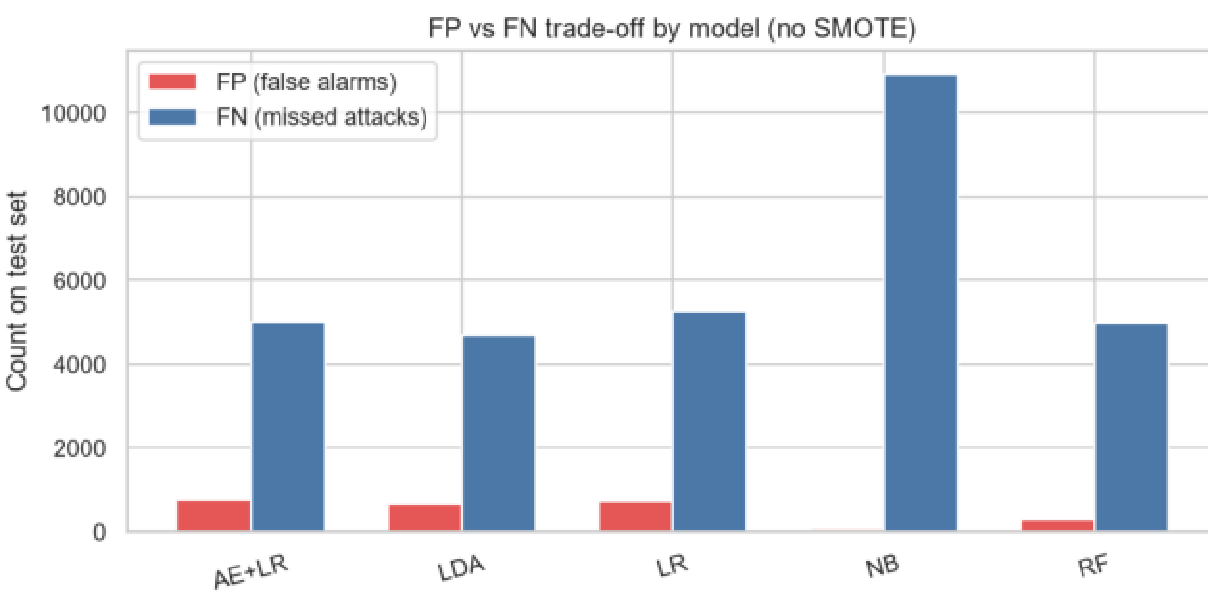
R2L and U2R remain poorly detected despite strong binary metrics.

Figure 8 — Confusion matrix (RF)



High TN/TP with substantial FN on minority attack patterns.

Figure 9 — FP/FN trade-off



Model profiles for SOC deployment: NB conservative, RF balanced.

6. Conclusions

Key findings. • Reproduced all author NSL-KDD metrics within tolerance using replication package. • AE+LR leads only in author CSV pipeline; LDA/RF competitive or superior under strict preprocessing. • SMOTE: modest aggregate F1 gains; does not fix rare-attack detection. • NB confirms precision-first profile unsuitable as standalone detector. • FAR and MCC required alongside accuracy for SOC-relevant comparison.

Strengths of source work. Public code, fixed seeds, FAR reporting, multiple baselines, official train/test split.

Weaknesses. Binary evaluation masks R2L/U2R failure; legacy dataset; hidden CSV ETL; encoding leakage risk; hybrid complexity poorly justified on F1.

Future work. Multiclass or hierarchical evaluation; contemporary datasets (CICIDS2017 subset); operational threshold calibration; SHAP/feature importance for interpretability.

7. Summing It Up

Arcos-Argudo et al. deliver a reproducible classical-ML benchmark with thoughtful metric reporting, but headline claims require qualification. AE+LR superiority (C1) holds only in their bundled preprocessing pipeline. SMOTE (C2), LDA/LR competitiveness (C4), NB behavior (C5), and FAR necessity (C6) are well supported. Leakage-free preprocessing (C3) fails under strict audit of author code. Byte-level reproducibility (C7) is partial — metrics yes, raw-data pipeline no.

Verdict: Conditionally recommend as a benchmarking reference for NSL-KDD tabular IDS studies. Do not deploy or cite near-0.90 AUC as operational readiness without per-attack-family evidence on modern traffic.

Repository: notebooks/ids_nsl_kdd_analysis.ipynb, docs/critical_evaluation.md, results/experiment_metrics.csv.